COMP0082 Coursework Report — Protein Enzyme Classification

March 2026

1 Introduction

Enzymes are biological catalysts that underpin virtually every metabolic process in living organisms. The Enzyme Commission (EC) numbering system classifies enzymes into six top-level classes— Oxidoreductases (EC 1), Transferases (EC 2), Hydrolases (EC 3), Lyases (EC 4), Isomerases (EC 5), and Ligases (EC 6)—according to the reaction they catalyse. Computational prediction of enzyme class from amino acid sequence alone is of substantial practical importance: it accelerates the functional annotation of the millions of uncharacterised proteins generated by high-throughput sequencing, and it supports drug target identification, metabolic engineering, and synthetic biology.

This project addresses the problem as a seven-class supervised classification task, where class 0 represents *Not an enzyme* and classes 1–6 correspond to the six EC classes. The dataset comprises 39,764 non-homologous protein sequences with a heavily skewed distribution: class 0 constitutes 81.5% of the data, while the smallest class (Ligase, EC 6) accounts for only 0.7%. This extreme class imbalance is the central challenge of the project and informs every design decision, from loss weighting and oversampling strategies to the choice of evaluation metrics.

Table 1: Dataset class distribution.

Class	Label	Count
0	Not an enzyme	32410
1	Oxidoreductase	1184
2	Transferase	2769
3	Hydrolase	2108
4	Lyase	600
5	Isomerase	411
6	Ligase	282

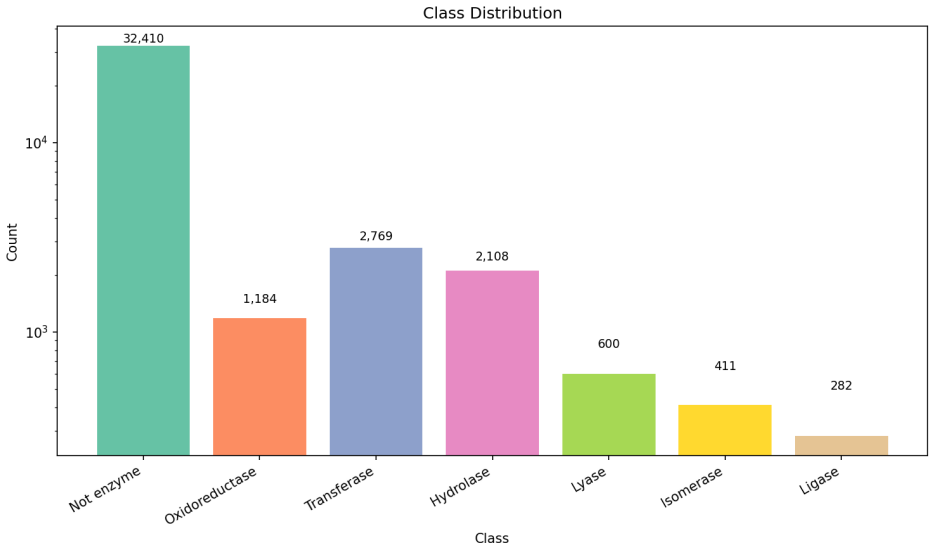


Figure 1: Observed class imbalance in the training dataset.

2 Methods

2.1 Feature Engineering

Amino acid composition (20-d). The normalised frequency of each canonical amino acid, capturing global residue distribution.

Dipeptide frequencies (400-d). Normalised counts of all ordered amino acid pairs, encoding local sequential context that single-residue composition cannot capture. Combined with a sequence-length scalar, this yields a 421-dimensional handcrafted composition vector.

Physicochemical properties (8-d). Molecular weight, isoelectric point, aromaticity, GRAVY hydropathicity, charge at pH 7.0, and Chou–Fasman secondary-structure propensity fractions (helix, turn, sheet), computed using BioPython ProtParam.

Protein language model embeddings. ESM-2 was used as a feature extractor. Two model sizes were evaluated: `esm2_t6_8M_UR50D` (320-d) and `esm2_t33_650M_UR50D` (1,280-d). Final-layer hidden states were mean-pooled across non-special tokens.

2.2 Experimental Design

All experiments used stratified 5-fold cross-validation (seed 42). Leakage prevention rules were enforced:

1. Fold indices generated before feature normalisation.
2. `StandardScaler.fit()` on training fold only.
3. SMOTE on training fold only.
4. Features derived only from amino acid sequence.
5. Precomputed ESM embeddings treated as stateless features; normalisation still fold-local.

Metrics: Accuracy, Macro F1, Balanced Accuracy, MCC. Macro F1 was the primary model-selection criterion.

2.3 Models

Baselines: Logistic Regression (balanced class weights) and Random Forest variants on hand-crafted features.

Advanced models: XGBoost and LightGBM on ESM-2 650M + physicochemical features (1,288-d), with and without SMOTE.

Fine-tuning: End-to-end ESM-2 8M fine-tuning with classification head $\text{Linear}(320, 128) \rightarrow \text{GELU} \rightarrow \text{Dropout}(0.3) \rightarrow \text{Linear}(128, 7)$, AdamW differential learning rates, OneCycleLR, mixed precision, gradient accumulation, early stopping.

Ensemble: Soft-vote blend of out-of-fold probabilities from fine-tuned 8M and XGBoost+SMOTE 650M. Weights were proportional to each model's OOF Macro F1.

3 Results

3.1 Model Comparison

Table 2 reports stratified 5-fold CV performance. The final Ensemble row uses the generated OOF blend metrics from `outputs/ensemble_results.json`.

Table 2: Model comparison (mean ± std for CV models; ensemble from OOF blend).

Model	Accuracy	Macro F1	Bal. Acc.	MCC
Logistic Regression	0.457 ± 0.003	0.191 ± 0.002	0.300 ± 0.010	0.171 ± 0.003
Random Forest	0.816 ± 0.001	0.141 ± 0.002	0.149 ± 0.001	0.093 ± 0.011
RF + PCA(50)	0.816 ± 0.002	0.197 ± 0.002	0.185 ± 0.001	0.216 ± 0.006
XGBoost	0.886 ± 0.003	0.575 ± 0.013	0.516 ± 0.011	0.621 ± 0.010
LightGBM	0.879 ± 0.003	0.515 ± 0.013	0.467 ± 0.008	0.605 ± 0.010
LightGBM + SMOTE	0.880 ± 0.002	0.576 ± 0.014	0.535 ± 0.010	0.620 ± 0.006
XGBoost + SMOTE	0.886 ± 0.002	0.595 ± 0.014	0.552 ± 0.011	0.631 ± 0.006
ESM-2 8M Fine-tune	0.823 ± 0.008	0.509 ± 0.013	0.570 ± 0.015	0.553 ± 0.008
Ensemble (OOF)	**0.890**	**0.625**	**0.605**	**0.659**

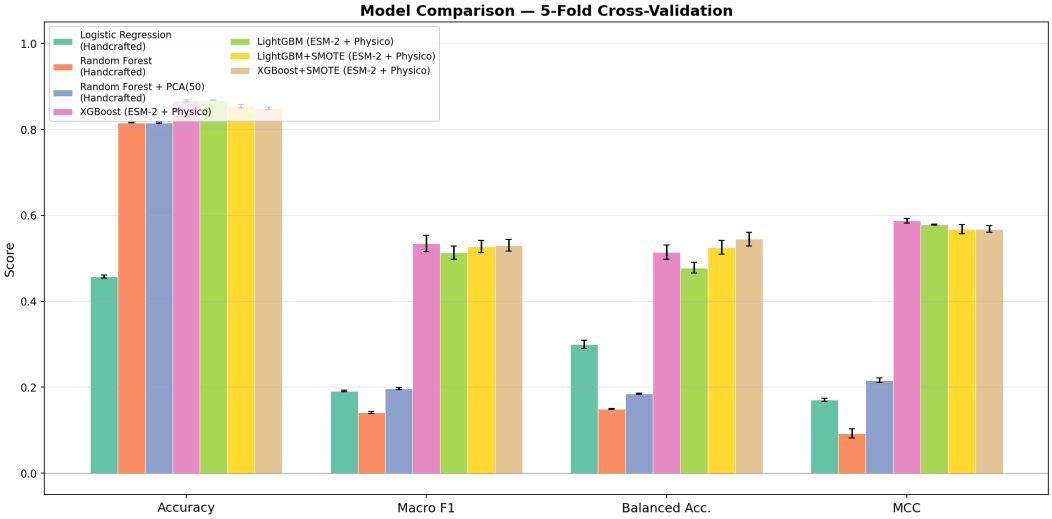


Figure 2: Aggregate comparison across major models.

3.2 Feature Ablation

Table 3: Feature ablation study (XGBoost, balanced weights).

Feature Set	Dimensions	Macro F1
Handcrafted only	429	0.206
ESM-2 only	1280	0.562
ESM-2 + Handcrafted	1709	0.552
ESM-2 + Physicochemical	1288	0.575

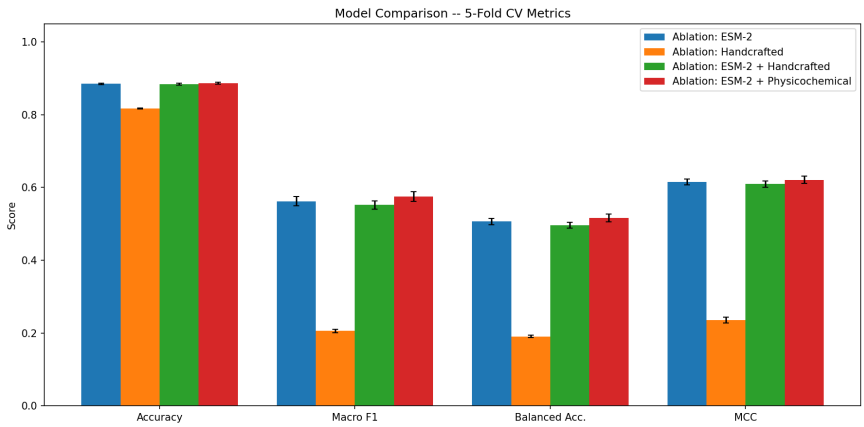
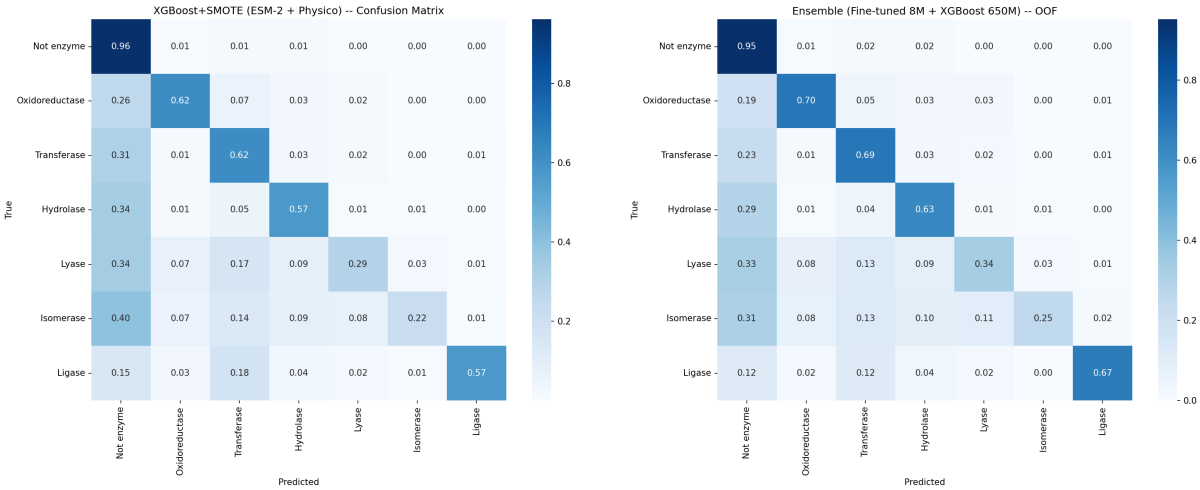


Figure 3: Ablation results showing dominant contribution of ESM-2 representations.

3.3 Confusion Matrix Analysis



(a) Best single model: XGBoost + SMOTE

(b) Ensemble OOF confusion matrix

Figure 4: Row-normalised confusion matrices for top-performing systems.

The best single model strongly predicts class 0 (Not enzyme), while Lyase and Isomerase remain challenging. The ensemble improves overall Macro F1 and Balanced Accuracy through complementary error patterns between the two constituent models.

3.4 Confidence Calibration

Table 4: Confidence calibration from `outputs/confidence_results.json`.

Level	Count (%)	Accuracy	Mean Max Prob
High ($p \geq 0.80$)	30448 (76.6%)	0.951	0.959
Medium ($0.50 \leq p < 0.80$)	6658 (16.7%)	0.662	0.663
Low ($p < 0.50$)	2658 (6.7%)	0.412	0.411

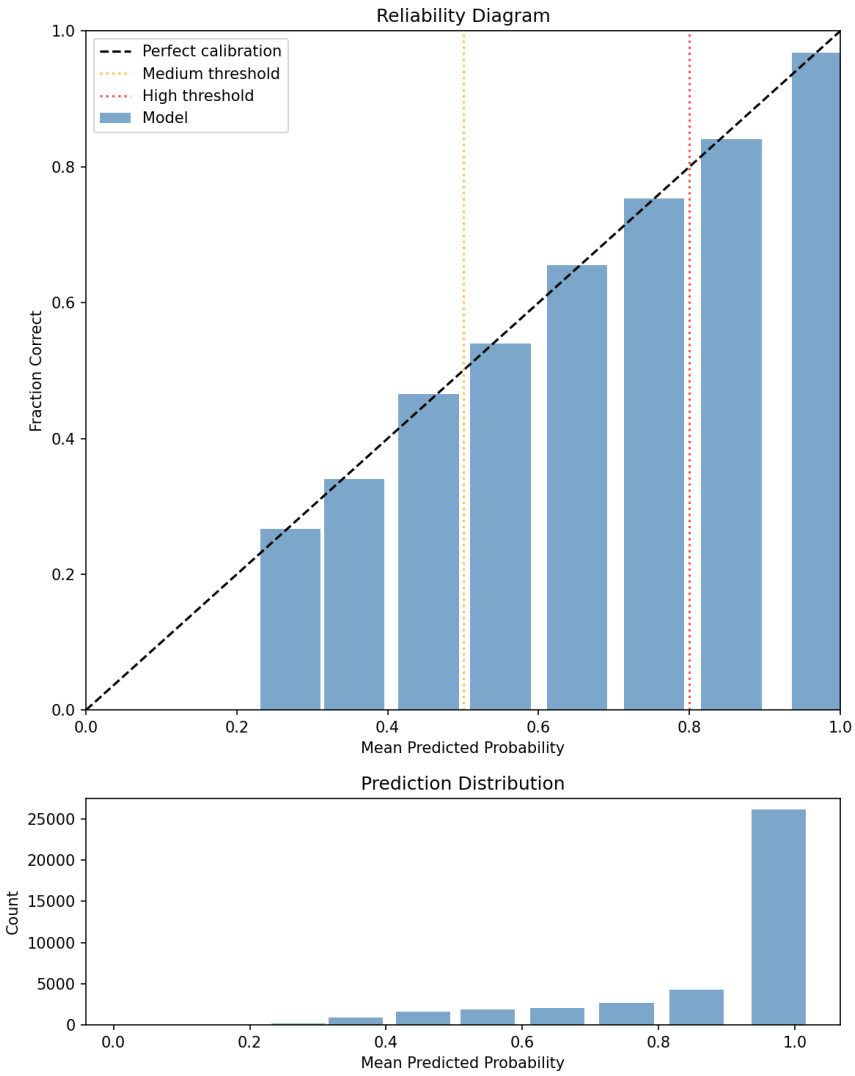


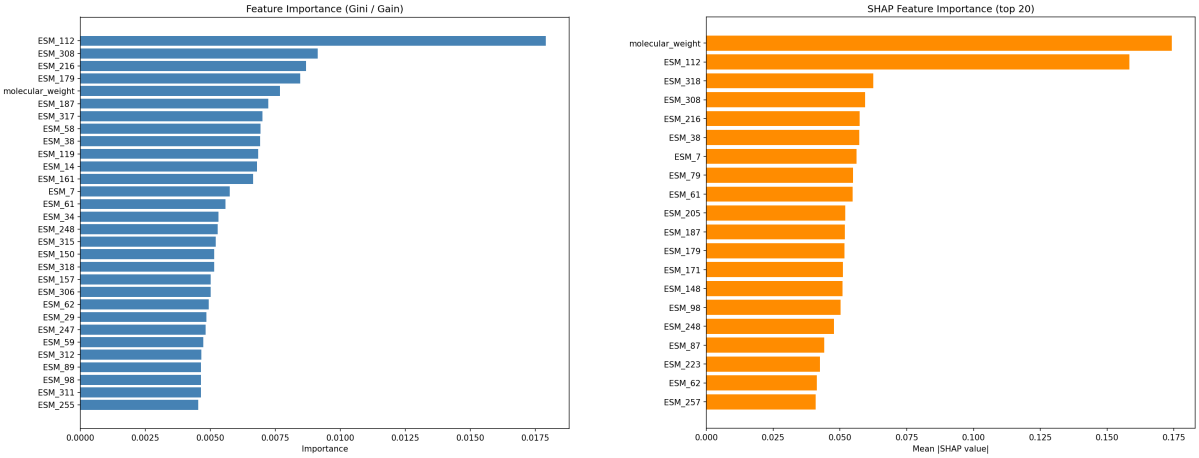
Figure 5: Reliability diagram for confidence calibration.

4 Interpretability

Interpretability was conducted on the best single model (XGBoost + SMOTE).

Feature importance. ESM dimensions dominate top features. The highest-ranked handcrafted feature is molecular weight.

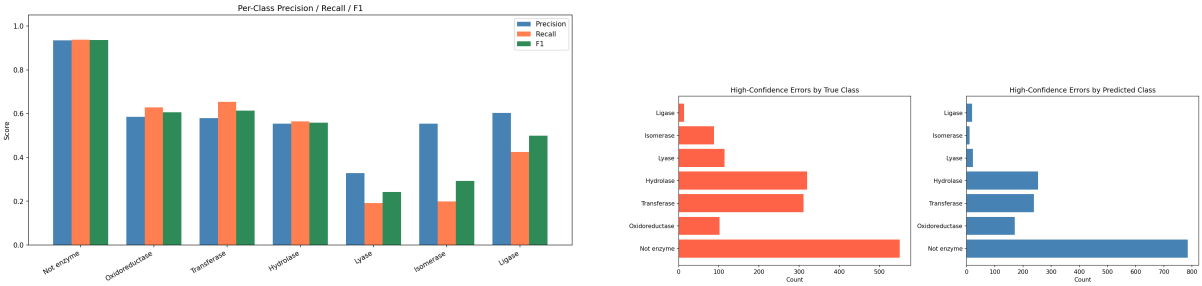
SHAP analysis. SHAP summary confirms strongest global effects from specific ESM dimensions.



(a) Gini feature importance

(b) SHAP importance

Figure 6: Complementary interpretability views of the best tree-based model.



(a) Per-class precision/recall/F1

(b) High-confidence error breakdown

Figure 7: Class-level behavior and error concentration at high confidence.

Table 5: Per-class metrics for XGBoost + SMOTE (OOF).

Class	Precision	Recall	F1
Not enzyme	0.934	0.938	0.936
Oxidoreductase	0.586	0.629	0.607
Transferase	0.579	0.653	0.614
Hydrolase	0.554	0.565	0.559
Lyase	0.329	0.192	0.242
Isomerase	0.554	0.200	0.293
Ligase	0.603	0.426	0.499

5 Discussion

5.1 What Worked

The largest gain came from ESM-2 650M embeddings with gradient-boosted trees, raising Macro F1 from roughly 0.19 to roughly 0.58–0.60. SMOTE provided a consistent additional lift on

minority classes. The soft-vote ensemble further improved Macro F1 to 0.6247 and MCC to 0.6586.

5.2 Limitations

Lyase and Isomerase remain difficult because of limited sample counts. Fine-tuned ESM-2 8M underperformed frozen 650M embedding-based XGBoost, likely due to model capacity gap and overfitting risk on the available dataset size. Adding the full 429-d handcrafted block to ESM-2 features was mildly detrimental.

5.3 Additional Generated Ensemble Data

From `outputs/ensemble_results.json`:

- Weight fine-tune: $w_{ft} = 0.4608$
- Weight XGBoost: $w_{xgb} = 0.5392$
- Plain blend Macro F1: 0.6246
- Optimised-threshold Macro F1: 0.6247
- Threshold vector approximately unity: $[1.0000, 0.9999, 0.9999, 0.9999, 1.0000, 0.9999, 0.9999]$

6 Conclusion

Protein language model representations dominate handcrafted sequence descriptors for this task. A robust leak-safe CV protocol, explicit imbalance handling, and model ensembling yielded strong performance on a severely imbalanced 7-class problem. The final ensemble achieved the best overall trade-off: **Accuracy 0.890, Macro F1 0.625, Balanced Accuracy 0.605, MCC 0.659**.

7 Appendix

7.1 Blind Challenge Prediction Command

```
python -m src.predict_blind \
    --fasta <blind_test.fasta> \
    --model outputs/models/best_model.joblib \
    --model-finetune outputs/models/finetune_artifact.joblib \
    --output outputs/predictions/blind_predictions.txt
```

Output format:

```
SEQ01 1 Confidence High
SEQ02 0 Confidence Medium
```

7.2 Software Dependencies

```
biopython>=1.83    numpy>=1.26     pandas>=2.1       scikit-learn>=1.4
xgboost>=2.0       lightgbm>=4.2   torch>=2.1        fair-esm>=2.0
```

```
matplotlib>=3.8    seaborn>=0.13   shap>=0.44        joblib>=1.3
scipy>=1.12        imbalanced-learn>=0.12             tqdm>=4.66
```

7.3 Reproducibility

All random seeds are fixed to 42 for `random`, `numpy`, and `torch`. Cross-validation uses stratified 5-fold splitting with shuffling and fixed random state. Result artifacts are persisted as JSON and reusable model files under `outputs/`.